

Samir Bhusal | Software QA Engineer | SDET

samir.masterprep@gmail.com | bhusalsamir.com.np | github.com/samirbhusal

Summary

QA Automation Engineer with 2+ years of experience designing and building BDD automation frameworks for web and mobile applications using Selenium, Appium, and Java. Currently pursuing an MS in Information Systems at Dakota State University and expanding automation engineering skills with Playwright, TypeScript, and API-driven testing.

Education

Dakota State University | Madison, SD

August 2025 – Present

Master of Science, Information Systems

Purbanchal University | Kathmandu, Nepal

October 2017 – October 2022

Bachelor of Science, Computer Applications

Experience

Software Engineer – Quality Assurance | Deerhold Ltd., Kathmandu, Nepal November 2022 – August 2025

- Maintained BDD automation framework for **Higi platforms** (web + mobile) using Selenium, Appium, and Java, achieving 95% regression automation coverage, 90%+ pass rate, and reducing release cycle time by 30%.
- Designed and maintained BDD test automation framework for **InStride Health** (mobile app – Android & iOS)) with 80% automation coverage using Selenium, Appium and Java.
- Executed API tests to prepare test data and preconditions for web and mobile automation execution.
- Led QA process for legacy system rebuilds, achieving production readiness in 3 months for **Wellness Portal**.
- Managed test planning, test case design, execution, defect tracking, and release documentation across web and mobile platforms.
- Collaborated in agile ceremonies ensuring alignment with cross-functional teams for continuous improvement.

Graduate Research Assistant | Dakota State University, Madison, SD

August 2025 – Present

- Designed AI helpdesk assistant **prototype** for incident-resolution workflows.
- Supported XAI healthcare research by maintaining literature matrix and evaluating Python analytics projects.
- Assisted in rubric design and grading for analytics coursework.

Projects

Python BDD Automation Framework | Python, Behave, Selenium, Page Object Model

- Developed a scalable Python Behave + Selenium POM framework with multi-browser & environment-driven execution.

Playwright BDD Automation Framework | TypeScript, Playwright, Cucumber

- Developed a scalable Playwright BDD automation framework with UI/API testing, multi-browser execution, BrowserStack integration, and Allure reporting.

Full Stack RAG Chatbot | React, Python, FastAPI, LangChain, Pinecone, Groq (Llama-3.3), HuggingFace

- Developed a RAG chatbot for intelligent incident query resolution using internal documents.

Technical Skills

Languages:	Java, Python, TypeScript, SQL, Gherkin
Test Automation:	Playwright, Selenium, Appium, Cucumber, Behave
Tools & Platforms:	Git, GitHub, JIRA, BrowserStack, Postman
Testing:	API, Regression, Smoke, UAT
Methodologies:	Agile/Scrum, SDLC, STLC

Achievements / Awards

- Winner of Hult Prize Purbanchal University 2020